2025 CSC493 – Capstone Weekly Reports[1]

<Download this template and name it Week_<WEEK_NUMBER>, where "<REPORT_NUMBER>" is replaced by the current week number (i.e., "1" for the first weekly report and so forth).

Week	Week 8
Project Name	Moodr AI
Report Date	Oct 14, 2025 ← click to select date

Part 1: Weekly Progress Report

- **Accomplishments**: **What did you complete this week?** Be specific and measurable.
- Examples: *Implemented login authentication using OAuth2; Fixed bug in file upload function (issues #14); Completed draft of UI wireframes; Identified functional and non-functional requirements; etc. (up to 3 points)*

This week, I have managed to get my tkinter GUI mostly completed. It is able to run the AI model, take a music file as input, run the model on the music file, and display an output on a colored graph segmented into quadrants that detail the mood each quadrant relates to. The only thing that my GUI is lacking is styling, as I have focused more on the displayed info and functionality so far.

- **Challenges**: **What problems did you encounter?** Mention blockers, unresolved issues, or risks.
- Examples: *Struggled with connecting database migrations to production server; API documentation was incomplete – need clarification; etc. (up to 3 points)*

The problem I have encountered, which has been an issue for the week, is the use of tkinter themes. For some reason, when I attempt to use a tkinter theme to enhance the appearance of my tkinter GUI window, it cannot find the PNGs required for the various parts of a tkinter UI. I have spent a considerable amount of time debugging this, and I believe it might be an issue that I can address later, as I can simply use the default theme for tkinter and keep it functional.

- **Next Steps**: **What are your goals for next week?** Keep them realistic and linked to project milestones.
- Examples: Set up automated testing with GitHub Actions; Final project proposal *(up to 2 points)*

My goals for this upcoming week are to implement a convolutional neural network model and link it to my GUI. I also plan on connecting the other two models that have yet to be connected to my GUI in order to allow the user of the program to choose the AI model they wish and use that model to classify the music of their choice by mood.

- **Discussion Points**: **What would you like to discuss in class, or with your professor/TA?** This could include technical issues, project direction, or collaboration concerns.
- Examples: *Should we prioritize front-end features before database scaling? Need guidance on writing integration tests; etc. (up to 2 points)*

Should I integrate the CNN model as a separate file, for clarity and readability, and load it alongside the logistic regression classifiers, or add the CNN model to my baseline_classifier file? Would it be helpful to expand the DEAM dataset with additional audio samples for CNN training, or should I stick with the current dataset for consistency across comparisons?

[1] Detailed Weekly Report requirements can be found here: 2024 URCPP Capstone - Using Agile and Reporting Out

Part 2: Time Reporting

Make sure that as you fill out the first prompt, you include in enough detail in the summary. For example, "debugging" is vague, but "debugged function X to make sure that when user does action Y, it is called and returns the value Z" is better.

- **Time Spent**: Briefly explain how much time, *outside of class*, spent on your project. If you worked on multiple components, each should get a detailed summary. Make sure to add up all the hours and minutes correctly. *Add as many rows as you need to the table below. Please do not include hours in class as part of this section. (up to 4 points)*

START	FINISH	HOURS	DETAILED SUMMARY
MM/DD HH:MM	MM/DD HH:MM		
10/7 08:00	10/7 12:00	4	I spent some time researching different tkinter styling techniques in order to best fit the output graph onto the UI and have it be both legible and easily understandable.
10/8 16:00	10/8 20:00	4	I implemented the color-coded mood quadrants onto the graph in order to allow for easy understanding of what mood each quadrant stands for.
10/9 08:00	10/9 10:00	2	I attempted to add a theme to my tkinter GUI file and ran into the roadblock mentioned earlier (the fact that the pngs from the theme folder would not be found by the theme tcl file)
10/13 16:00	10/13 20:00	4	I again attempted to fix the issue with the theme and could not fully fix it. I instead decided to add an option to revert the theme back to default if the custom theme could not be used.
	WEEKLY TOTAL	14	

- **Total (Cumulative) Project Time Spent**: After the number of hours and minutes, make sure to briefly explain whether you are on track and if not, what you may need to do in order to achieve what you set out to accomplish. *(up to 2 points)*

So far, I have spent 108 hours on this project so far. I am technically a little behind on the schedule created using the help of ChatGPT, but I believe that I can pivot the issue of using themes from the GUI goals to a stretch goal and possibly solve that issue further down the line as it is just a issue related to the visuals and not the core functionality of the program. I need to add a CNN model this week, connect it to the GUI, and allow for choice of model on the GUI.

Rubric:

The following rubric will be used, but they might change as needed.

Accomplishments (3 points)

1 point for a general description of progress, 2 points for specifics on progress, 3 points for specifics AND referring to previous targets and explaining how current accomplishments build on previous ones.

Challenges (3 points)

1 point for mentioning there are roadblocks, 2 points for specifics, 3 points for specifics AND what was done already to try to overcome them.

Desired discussion points (2 points)

1 point for at least one relevant discussion point as a general question, 2 points for relevant discussion points with specifics

Future Goals (2 points)

1 point for concrete future targets (i.e. "working more on the project" is a zero, but "working on getting component X to interface with component Y" suffices), 2 points for tying in the targets with what was hopefully discussed in the meeting.

Time Spent (4 points)

1 point for including general statements of how much time was spent ("4 hours on coding"), 2 points for splitting time into specific parts ("1.5 hours on research on component X, 1 hour coding, 2.5 hours debugging"), 3 points for specific parts and details on the pieces ("1.5 hours researching Turtle interface for drawing concentric circles given inputs from the user, 1 hour coding function X that used that interface, 2.5 hours testing function X by giving it multiple values and fixing errors for values A, B, C, and D"). 1 Point for totalling the hours correctly.

What happens if your time on a task is interrupted and you don't have a concrete (or discrete) end time? In this case put the start time in, and the word "interrupted" for the end time and include the task total time. Rounding to the nearest 15 minutes is acceptable. *(This makes adding up times easier, especially when you use decimal hours, i.e. 3.75 rather than 3 hours 45 minutes.)*

Total (Cumulative) Project Time (2 points)

1 point for summing the values correctly, 2 points for the total time AND reflection on progress (you are confident to fit the target and if not, what course corrections you anticipate needing to make)

Resources

Here's a link to this Weekly Report Template – Make a copy and use it:

2025 CSC493 Weekly Report v2 TEMPLATE